

## NIIGATA, Japan

WMO: 476040

Lat: 37.8936N

Lon: 139.0189E

Elev: 19

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 28.9       | 30.2 | 18.1                          | 13.7 | 34.6 | 19.8 | 14.9 | 35.2 | 25.9                  | 38.6 | 24.1 | 37.3 | 9.7                   | 300  | 0.430 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 11.3                   | 91.4            | 77.4 | 89.1 | 76.8 | 86.7 | 76.1 | 79.1                | 87.4 | 78.1 | 86.2 | 77.1 | 85.0 | 8.8                  | 250  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 76.7                            | 139.2 | 83.9 | 75.6 | 134.3 | 83.3 | 74.6 | 129.7 | 82.5 | 42.6          | 87.6 | 41.6 | 86.3 | 40.6 | 85.3 | 82.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 22.3              | 19.4 | 16.8 | DB                         | 25.8 | 96.5               | 1.8 | 2.1                                                | 24.5 | 98.0       | 23.5 | 99.3       | 22.4 | 100.5      | 21.1 | 102.0 |
|                   |      |      | WB                         | 23.8 | 80.6               | 1.7 | 1.1                                                | 22.6 | 81.4       | 21.6 | 82.0       | 20.7 | 82.6       | 19.4 | 83.4  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 57.7   | 37.2  | 37.7  | 43.2  | 53.0  | 63.1  | 70.1  | 77.6  | 80.5  | 73.3  | 62.5  | 51.3  | 42.0 |
|                                                                                  | DBStd       | 15.82  | 3.80  | 4.41  | 5.25  | 6.16  | 5.48  | 4.13  | 4.64  | 4.34  | 5.41  | 5.13  | 5.56  | 4.79 |
|                                                                                  | HDD50       | 1302   | 397   | 346   | 221   | 37    | 0     | 0     | 0     | 0     | 0     | 0     | 50    | 251  |
|                                                                                  | HDD65       | 4012   | 862   | 766   | 677   | 364   | 102   | 7     | 0     | 0     | 2     | 110   | 410   | 712  |
|                                                                                  | CDD50       | 4132   | 0     | 1     | 10    | 126   | 405   | 604   | 856   | 947   | 699   | 389   | 91    | 4    |
|                                                                                  | CDD65       | 1362   | 0     | 0     | 0     | 3     | 42    | 160   | 391   | 482   | 251   | 33    | 0     | 0    |
|                                                                                  | CDH74       | 10273  | 0     | 0     | 0     | 16    | 173   | 528   | 3080  | 4863  | 1543  | 70    | 0     | 0    |
|                                                                                  | CDH80       | 3181   | 0     | 0     | 0     | 1     | 20    | 71    | 954   | 1734  | 393   | 8     | 0     | 0    |
| Wind                                                                             | WSAvg       | 7.5    | 9.3   | 8.6   | 8.2   | 7.8   | 7.1   | 6.4   | 6.2   | 6.3   | 6.6   | 6.7   | 7.5   | 9.0  |
| Precipitation                                                                    | PrecAvg     | 71.5   | 7.4   | 5.0   | 4.2   | 3.8   | 3.9   | 4.6   | 7.5   | 5.7   | 6.5   | 6.1   | 7.8   | 9.1  |
|                                                                                  | PrecMax     | 94.4   | 14.2  | 10.3  | 7.2   | 5.9   | 8.5   | 14.9  | 17.4  | 24.3  | 13.2  | 11.2  | 16.4  | 17.5 |
|                                                                                  | PrecMin     | 57.8   | 4.2   | 2.2   | 1.7   | 1.3   | 1.4   | 1.5   | 0.6   | 0.2   | 1.0   | 0.6   | 3.3   | 3.9  |
|                                                                                  | PrecStd     | 9.4    | 2.2   | 1.6   | 1.1   | 1.1   | 1.7   | 2.7   | 4.4   | 4.5   | 2.9   | 2.0   | 2.8   | 3.0  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 51.7  | 56.3  | 65.7  | 76.3  | 82.5  | 86.2  | 93.7  | 95.2  | 91.9  | 79.6  | 69.6  | 59.4 |
|                                                                                  |             | MCWB   | 45.6  | 46.8  | 51.3  | 58.3  | 65.0  | 70.9  | 77.2  | 77.6  | 76.3  | 67.5  | 58.6  | 52.2 |
|                                                                                  | 2%          | DB     | 47.8  | 50.9  | 60.0  | 70.8  | 78.5  | 81.8  | 90.5  | 92.1  | 87.7  | 75.5  | 65.3  | 55.1 |
|                                                                                  |             | MCWB   | 42.6  | 44.7  | 49.0  | 56.2  | 62.6  | 69.1  | 77.0  | 77.8  | 75.1  | 65.8  | 56.9  | 49.6 |
|                                                                                  | 5%          | DB     | 45.3  | 47.6  | 55.4  | 66.7  | 75.3  | 79.2  | 88.0  | 90.0  | 84.4  | 73.1  | 62.5  | 52.3 |
|                                                                                  |             | MCWB   | 40.4  | 41.7  | 47.1  | 54.6  | 61.6  | 68.4  | 76.6  | 77.5  | 73.8  | 64.4  | 55.8  | 47.0 |
|                                                                                  | 10%         | DB     | 43.3  | 44.9  | 52.0  | 63.0  | 72.2  | 76.9  | 85.3  | 87.9  | 81.5  | 70.9  | 60.1  | 49.7 |
|                                                                                  |             | MCWB   | 38.8  | 39.8  | 45.6  | 53.1  | 60.8  | 67.5  | 75.7  | 77.2  | 72.5  | 62.8  | 54.1  | 44.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 47.3  | 48.9  | 53.4  | 61.9  | 67.8  | 73.9  | 79.6  | 80.8  | 78.4  | 70.9  | 61.3  | 54.1 |
|                                                                                  |             | MCDB   | 50.3  | 54.1  | 61.2  | 68.4  | 75.8  | 79.6  | 89.0  | 89.4  | 87.3  | 76.3  | 65.2  | 57.7 |
|                                                                                  | 2%          | WB     | 43.4  | 45.2  | 50.6  | 58.3  | 65.5  | 72.0  | 78.3  | 79.6  | 76.2  | 68.2  | 58.9  | 50.2 |
|                                                                                  |             | MCDB   | 46.8  | 49.8  | 57.1  | 67.0  | 73.9  | 77.5  | 87.3  | 87.7  | 84.9  | 73.2  | 63.7  | 54.9 |
|                                                                                  | 5%          | WB     | 41.3  | 42.5  | 48.3  | 56.2  | 63.9  | 70.7  | 77.3  | 78.8  | 74.8  | 66.0  | 56.7  | 47.6 |
|                                                                                  |             | MCDB   | 44.5  | 47.0  | 54.1  | 63.8  | 71.9  | 76.4  | 85.5  | 86.6  | 82.5  | 71.3  | 61.2  | 51.5 |
|                                                                                  | 10%         | WB     | 39.4  | 40.2  | 46.2  | 54.4  | 62.2  | 69.3  | 76.3  | 77.9  | 73.6  | 63.9  | 54.8  | 45.4 |
|                                                                                  |             | MCDB   | 42.8  | 44.5  | 51.8  | 61.5  | 69.8  | 75.2  | 83.8  | 85.7  | 80.3  | 69.5  | 59.5  | 49.1 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.3   | 9.8   | 12.4  | 14.1  | 13.8  | 11.1  | 10.5  | 11.3  | 11.4  | 12.3  | 11.7  | 9.6  |
|                                                                                  |             | MCDBR  | 12.0  | 15.0  | 19.9  | 20.9  | 18.8  | 14.5  | 14.2  | 14.0  | 14.3  | 14.5  | 15.8  | 13.6 |
|                                                                                  |             | MCWBR  | 9.2   | 11.0  | 12.8  | 10.5  | 8.5   | 6.5   | 5.8   | 5.8   | 6.6   | 7.9   | 10.5  | 10.7 |
|                                                                                  | 5% WB       | MCDBR  | 11.6  | 14.6  | 16.8  | 16.4  | 15.4  | 11.7  | 12.7  | 12.5  | 12.6  | 11.3  | 13.5  | 12.7 |
|                                                                                  |             | MCWBR  | 9.9   | 11.6  | 12.2  | 9.6   | 8.4   | 6.0   | 5.5   | 5.7   | 6.1   | 7.6   | 10.6  | 10.9 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.420  | 0.455 | 0.517 | 0.550 | 0.556 | 0.561 | 0.569 | 0.545 | 0.484 | 0.448 | 0.432 | 0.426 |      |
|                                                                                  | taud        | 1.992  | 1.910 | 1.832 | 1.814 | 1.864 | 1.932 | 1.972 | 2.042 | 2.169 | 2.154 | 2.100 | 2.017 |      |
|                                                                                  | Ebn at Noon | 226    | 236   | 234   | 236   | 238   | 236   | 233   | 235   | 243   | 238   | 222   | 213   |      |
|                                                                                  | Edh at Noon | 42     | 51    | 61    | 66    | 64    | 60    | 57    | 52    | 43    | 40    | 37    | 38    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 443    | 703   | 1084  | 1435  | 1657  | 1635  | 1489  | 1558  | 1235  | 923   | 581   | 400   |      |
|                                                                                  | RadStd      | 44     | 75    | 97    | 135   | 173   | 126   | 212   | 169   | 94    | 71    | 47    | 48    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (18 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | +0.52 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page